

A singleton-event counterexample to the stated quadratic divergence law

Conjecture 00000007862 — literal displayed-bound clause

Scope. The bilingual source gives the bound

$$B(\Phi) = \frac{\sum_i p_i}{(1 - ep^*) \prod_i (1 - p_i)}$$

and then states that *the bound* diverges quadratically as $ep^* \rightarrow 1$. We disprove this unqualified law on a fixed admissible family. The proof concerns the displayed bound, without assuming or formalizing a formula for the expected restart count $T(\Phi)$.

Concrete probability and event model. There is one variable with two possible outcomes, 0 and 1. For $0 < p < 1$, put

$$\mathbb{P}(0) = p, \quad \mathbb{P}(1) = 1 - p.$$

These masses are nonnegative and sum to one. There is one bad event, $A = \{0\}$. Its probability, computed by summing the masses of its outcomes, is $\mathbb{P}(A) = p$. Thus the largest event probability is actually $p^* = p$, attained by the sole event.

The event depends on the sole variable. Define dependency adjacency by requiring distinct events whose variable supports overlap. This graph has no edges and degree $d = 0$: a singleton set of events contains no distinct pair. The usual symmetric LLL admissibility condition $ep^*(d+1) < 1$ therefore reduces to $ep < 1$. Any requirement of independence for distinct nonadjacent events is vacuous in this genuine singleton event family.

Approach to the critical value. Let $e = \exp(1) > 1$, and for every integer $n \geq 0$ choose

$$p_n = \frac{1 - 1/(n+2)}{e}, \quad \delta_n = 1 - ep_n = \frac{1}{n+2}.$$

Then $0 < p_n < 1/e < 1$, $ep_n < 1$, and $ep_n \rightarrow 1$ from below. This is an admissible sequence of actual probability laws on the same variable-event hypergraph.

For this family the finite sum and product in the source reduce to $\sum_i p_i = p_n$ and $\prod_i (1 - p_i) = 1 - p_n$. Consequently its *actual displayed bound* is the positive number

$$B_n = \frac{p_n}{\delta_n(1 - p_n)}.$$

No scalar probability or maximum is added as an unproved premise: both are computed from the concrete event family.

The asymptotic contradiction and its formal scope

First-order behavior. Multiplication by the critical gap gives

$$\delta_n B_n = \frac{p_n}{1 - p_n} \longrightarrow \frac{1/e}{1 - 1/e} = \frac{1}{e - 1} > 0.$$

Hence $B_n \sim (e - 1)^{-1} \delta_n^{-1}$. The displayed bound really diverges along this family, but its pole has order one.

Quadratic normalization. A second multiplication gives

$$\delta_n^2 B_n = \delta_n \frac{p_n}{1 - p_n} \longrightarrow 0.$$

For any fixed $c > 0$, convergence to zero implies that eventually $\delta_n^2 B_n < c$. Since $\delta_n > 0$, it follows that

$$\neg(\text{eventually } c \delta_n^{-2} \leq B_n).$$

Thus the bound cannot have a positive quadratic asymptotic coefficient. It also cannot satisfy a two-sided $\Theta(\delta_n^{-2})$ law, because such a law would supply the positive eventual lower comparison just excluded. This refutes the asserted universal quadratic divergence conjunct of the source.

What is, and is not, concluded. The source does not specify a worst-case supremum over instances or a growing family of event counts. An expression can have different behavior if its remaining factors also vary singularly in a specially chosen growing family. Such an existential or worst-case statement is a different claim, and is not disproved here. Our counterexample applies to the unqualified law when it is asserted for admissible families. This interpretation is explicit and must receive independent source-bound review.

The source’s other claims concern an upper bound for T , a branching process explanation, and an adaptive-selection exponent. We neither prove nor individually disprove those claims. A false necessary conjunct suffices to reject the original conjunction under the stated universal reading. No resampling algorithm is silently supplied to repair the source’s omitted definition of restart rounds.

Lean correspondence. `Main.lean` defines a finite probability law with nonnegative masses and total mass one; `instanceLaw`, `variableValue`, and `eventFamily` give the actual variable and event model. `pStar_largest` proves the maximum by `IsGreatest`; `dependencyDegree_zero` and `instance_admissible` establish degree-zero admissibility. `actualBound` retains the original finite expression. `actual_scaled_square_tendsto_zero`, `actual_scaled_linear_tendsto`, and `actual_no_positive_quadratic_lower_bound` prove the limits and contradiction for this actual bound. The premise-free theorem `literal_bound_counterexample` collects the certificate.

Verification status. The pinned Lean package and this proof are author artifacts. Deterministic replay, independent semantic review, and acceptance belong to the manager; this document does not self-certify them.